

Persuasive Ad Explainer

Problem: Older adults and everyday internet users are often exposed to emotionally persuasive online advertisements that use urgency, fear, scarcity, social proof, or exaggerated claims to influence purchasing decisions without users recognizing the tactic being used.

Solution: Our tool is designed to analyze the language current adverts use in order to determine if it is using an emotionally persuasive tactic.

Why us: Our group combines real-world ad data collected from social media platforms with NLP and vision-language models to create an explainable, ethically focused system for analyzing persuasive advertising language, while maintaining a third-party, unbiased approach that avoids misleading classifications of ads as scams or false advertising.

MVP: A user can analyze an advertisement and receive transparent feedback if that ad is using manipulative/persuasive language.

Direction of our Project-

We are exploring an NLP tool that explains emotionally persuasive advertising tactics. The tool would analyze online ad text, highlight phrases that use persuasion strategies, and explain the tactic in plain language.

We are not trying to legally classify ads as scams or false advertising. Instead, we are focusing on helping users recognize persuasive language such as urgency, FOMO, fear appeals, scarcity, exaggerated claims, authority manipulation, and social proof.

1.The Persona

Our target user is older adults, especially users age 60+, who regularly use Facebook, Instagram, YouTube, or news websites and encounter online ads for health products, supplements, insurance, financial services, or limited-time promotional offers.

Although emotionally persuasive ads can affect users of any age, we are starting with older adults because they are a more specific and potentially vulnerable user group. They may be more likely to encounter ads related to health, finance, insurance, or wellness products, and they may not always have an easy way to verify whether an ad is trustworthy before clicking or purchasing.

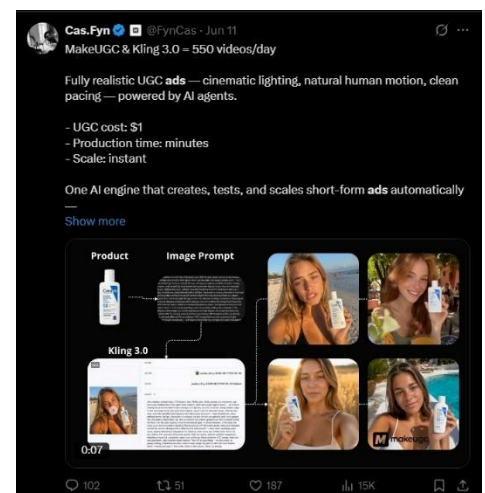
Over time, this tool could also help younger users or general consumers who have been influenced by urgency, FOMO, influencer marketing, exaggerated claims, or misleading promotional language.

2. The Pain

Many online ads use emotionally persuasive language such as urgency, FOMO, fear appeals, scarcity, exaggerated promises, social proof, or authority-based claims to pressure users into clicking, purchasing, or sharing personal information.

This becomes more concerning because ads on platforms like Instagram, Facebook, YouTube, and news websites are often highly personalized based on a user's online activity, interests, searches, clicks, engagement, and inferred preferences. For example, if a user has recently searched for travel information or interacted with travel-related content, they may start seeing ads for flights, hotels, resorts, or vacation packages. Because the ad feels personally relevant, the user may be more likely to trust it or respond emotionally.

The exact problem is that users often do not understand which parts of an ad are trying to emotionally pressure them or why the wording may deserve caution. As a result, they may make impulse purchases or trust misleading promotional claims without fully evaluating the product, source, or persuasive tactic being used.





We collected examples of ads and sponsored posts from X, Instagram, YouTube, and other websites. Many of these ads use tactics such as large discounts, limited time offers, exclusivity, FOMO, emotional pressure, or promises of high financial/personal benefit.

These ads are not necessarily false or illegal, but they can make an offer feel more urgent, profitable, exclusive, or personally relevant than it may actually be. The problem is that users often notice the offer, but not the persuasive tactic behind it. Our goal is to help users recognize those tactics before they click, purchase, download, or share personal information.

3. The Hack

Today, users usually deal with suspicious or persuasive ads in a manual and inconsistent way. Some users ignore the ad if it feels suspicious, while others click

through and try to judge the product after seeing the landing page. They may Google the product or company, check Amazon reviews, search Reddit threads, read comments under the ad, ask friends or family members, or report the ad to the platform.

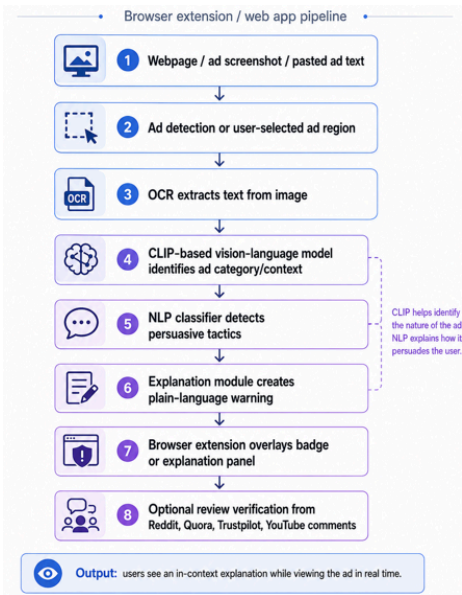
More recently, some users may also upload a screenshot or copy the ad text into tools like ChatGPT or Claude and ask whether the ad looks suspicious.

Existing research has explored related problems, such as detecting persuasion techniques in text and images [1], identifying persuasion strategies in advertisements [2], detecting dark patterns in web/mobile interfaces [3], and detecting persuasion in social media or political advertising [4]. However, these systems are mostly research-focused and do not directly give everyday users an immediate, plain-language explanation while they are viewing an ad.

How Our System Would Analyze Them

Example	Persuasive Tactic Detected	Why It Matters	Possible App Explanation
Kalshi soccer ad: "History, again?" and "\$100 → \$1,056"	Financial upside, sports emotion, FOMO	Connects betting/trading behavior to fandom and a large possible payout	This ad uses sports emotion and a large potential return to make the opportunity feel exciting. Users should understand the financial risk before acting.
Prop trading firm ad: "Exclusive offers"	Exclusivity, money opportunity	Makes the user feel they may miss a special financial opportunity	This ad uses exclusivity to make the offer feel limited or special. Users may feel pressure to click before comparing alternatives.
"You are COOKED if you can't make \$20K+ per month"	Fear/shame, unrealistic income promise	Pressures users by implying they are falling behind financially	This ad uses fear and shame to create urgency around making money online. The income claim may need verification before trusting it.
Hims ad: "Work hard, play hard" / "Help increase your sex drive"	Identity, insecurity, health/sexual confidence	Uses lifestyle messaging around sensitive personal concerns	This ad connects the product to confidence and lifestyle. Because it involves health-related decisions, users should review medical information carefully.
Botox ad: "Get \$65 off" / "Check if you're eligible now" / "Limited-time offer"	Discount, urgency, eligibility framing	Encourages quick action around a medical/cosmetic decision	This ad combines a discount with time pressure and eligibility language. Users may feel encouraged to act quickly before fully evaluating the decision.

The above table was created from the ad examples we collected across platforms like Instagram, X, and YouTube. We used these ads as input to a CLIP-based vision-language pipeline [5] to help identify the general nature of each ad.



User Our system analyzes online ads using a multimodal workflow. OCR extracts text from the ad, CLIP identifies the ad’s visual context, and an NLP classifier detects persuasive tactics. The final output is a real-time browser-extension warning that explains why the ad may be emotionally persuasive or misleading, with optional review verification from public discussion sources.

Our users, however, are not the only ones who have been affected by this problem. In fact, we discussed the problem informally with older friends and family members and also reflected on our own daily experience with online ads. The main insight is that people regularly encounter ads that feel urgent, personalized, or emotionally persuasive. Older users may be especially vulnerable because they are still adapting to modern digital platforms and may not always recognize when ads are using urgency, discounts, fear, or exclusivity to influence them. We also see this problem in our own generation through financial, health, AI tool, and lifestyle ads. The key learning is that users do not just need a warning that an ad is “bad”; they need a clear explanation of what tactic is being used and why they should pause before clicking, purchasing, downloading, or sharing personal information.

The Verdict

Based on our initial customer discovery and the ad examples we collected, we are proceeding with this idea. We believe emotionally persuasive ads are a real problem.

For now, our focus is a practical MVP: detecting persuasive tactics in ad text or screenshots and generating a clear explanation for the user. As the project develops, we may expand toward more advanced real-time ad monitoring, such as a Chrome extension that automatically detects sponsored content, overlays warning badges directly on ads, tracks repeated persuasive patterns across websites, and provide review verification from external sources.

As an advanced direction, we want to build a real-time persuasive ad monitoring extension that detects repeated emotional tactics across sponsored content and verifies ad claims against external signals such as reviews, public complaints, user feedback, and source credibility before the user clicks or purchases.

References

- [1] [SemEval-2021 Task 6: Detection of Persuasion Techniques in Texts and Images](#)
- [2] [Persuasion Strategies in Advertisements](#)
- [3] [Automated Detection of Dark Patterns in Mobile Applications / dark-pattern detection work](#)
- [4] [Spotting Persuasion: A Low-cost Model for Persuasion Detection in Political Ads on Social Media](#)
- [5] [Learning Transferable Visual Models From Natural Language Supervision](#)